

Lab Assignment :18_05

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Batch-02

Task 1 – Weather API Integration

Connect to a Weather API to fetch current weather details for a given city.

Instructions:

- Prompt AI to generate Python code using the requests library.
- Handle errors: invalid city name, API key errors, and network failures.
- Display temperature, humidity, and weather description in a readable format.

Expected Output:

- A working Python script that fetches and displays weather data with proper error handling.

Screenshots:

```

170
171 """
172 import requests
173 def get_weather(city):
    '''get weather of city'''
    api_key = 'demo'
    url = f'https://api.openweathermap.org/data/2.5/weather?q={city}&appid={api_key}&units=metric'
    try:
        response = requests.get(url)
        response.raise_for_status() # Check for HTTP errors
        data = response.json()
        if data.get('cod') != 200:
            raise ValueError(f"City '{city}' not found.")
        return {
            'temperature': data['main']['temp'],
            'humidity': data['main']['humidity'],
            'description': data['weather'][0]['description']
        }
    except requests.exceptions.HTTPError as http_err:
        print(f"HTTP error occurred: {http_err}")
    except requests.exceptions.ConnectionError as conn_err:
        print(f"Connection error occurred: {conn_err}")
    except requests.exceptions.Timeout as timeout_err:
        print(f"Timeout error occurred: {timeout_err}")
    except ValueError as val_err:
        print(val_err)
    except Exception as err:
        print(f"An error occurred: {err}")

```

```

PS C:\Users\arell\Music\aiac> python -u "c:\Users\arell\Music\aiac\assg_18.py"
Enter the city name: london

Current weather in london:
Temperature: 12°C
Humidity: 88%
Description: Light rain, light drizzle and rain
PS C:\Users\arell\Music\aiac>

```

Task 2 – Currency Exchange Rates

Integrate a Currency Exchange API to convert INR to USD, EUR, and GBP.

Instructions:

- Use AI to generate API request code.
- Validate user input for amount and currency codes.
- Implement error handling for invalid responses and API downtime.
- Show conversion results clearly in tabular format.

Expected Output:

- A script that converts INR to multiple currencies with graceful

error handling.

Screenshots:

```
221
222 import requests
223 def convert_currency(amount_inr):
224     '''convert currency'''
225     url = 'https://api.exchangerate-api.com/v4/latest/INR'
226     try:
227         response = requests.get(url)
228         response.raise_for_status() # Check for HTTP errors
229         data = response.json()
230         rates = data.get('rates', {})
231         if not rates:
232             raise ValueError("Currency rates not found.")
233         conversion_results = {
234             'USD': amount_inr * rates.get('USD', 0),
235             'EUR': amount_inr * rates.get('EUR', 0),
236             'GBP': amount_inr * rates.get('GBP', 0)
237         }
238         return conversion_results
239     except requests.exceptions.HTTPError as http_err:
240         print(f"HTTP error occurred: {http_err}")
241     except requests.exceptions.ConnectionError as conn_err:
242         print(f"Connection error occurred: {conn_err}")
243     except requests.exceptions.Timeout as timeout_err:
244         print(f"Timeout error occurred: {timeout_err}")
245     except ValueError as val_err:
246         print(val_err)
247     except Exception as err:
248         print(f"An error occurred: {err}")
249 def main():
250     '''main function'''
251     try:
252         amount_inr = float(input("Enter the amount in INR: "))
253         if amount_inr < 0:
254             raise ValueError("Amount cannot be negative.")
255         conversion_results = convert_currency(amount_inr)
256         if conversion_results:
257             print("\nConversion Results:")
258             print(f"{'Currency':<10} {'Amount':>10}")
259             print("-" * 22)
260             for currency, amount in conversion_results.items():
261                 print(f"{'currency':<10} {'amount':>10.2f}")
262     except ValueError as val_err:
263         print(val_err)
264     except Exception as err:
265         print(f"An error occurred: {err}")
266 if __name__ == "__main__": main()
267
```

```
Currency      Amount
PS C:\Users\arell\Music\aiac> python -u "c:\Users\arell\Music\aiac\assg_18.py"
Enter the amount in INR: 2300

Conversion Results:
Currency      Amount
-----
Conversion Results:
Currency      Amount
-----
USD            24.38
EUR            21.16
USD            24.38
EUR            21.16
GBP            18.31
PS C:\Users\arell\Music\aiac> 
```

Task 3 – GitHub Repository Info Fetcher

Connect to the GitHub API to fetch details of a repository (e.g., stars, forks, issues).

Instructions:

- Prompt AI to write a Python function to send GET requests to GitHub API.
- Handle rate limit errors, 404 (repo not found), and invalid input.
- Display repository name, description, stars, forks, and open issues.

Expected Output:

- A Python program that retrieves and displays GitHub repository information with robust error handling.

Screenshots:

```

282 import requests
283 def get_repo_info(owner, repo):
284     '''get repository information'''
285     url = f'https://api.github.com/repos/{owner}/{repo}'
286     try:
287         response = requests.get(url)
288         response.raise_for_status() # Check for HTTP errors
289         data = response.json()
290         return {
291             'name': data.get('name', 'N/A'),
292             'description': data.get('description', 'N/A'),
293             'stars': data.get('stargazers_count', 'N/A'),
294             'forks': data.get('forks_count', 'N/A'),
295             'open_issues': data.get('open_issues_count', 'N/A')
296         }
297     except requests.exceptions.HTTPError as http_err:
298         if response.status_code == 404:
299             print(f"Repository '{owner}/{repo}' not found.")
300         elif response.status_code == 403 and 'X-RateLimit-Remaining' in response.headers and response.headers['X-RateLimit-Remaining'] == '0':
301             print("Rate limit exceeded. Please try again later.")
302         else:
303             print(f"HTTP error occurred: {http_err}")
304     except requests.exceptions.ConnectionError as conn_err:
305         print(f"Connection error occurred: {conn_err}")
306     except requests.exceptions.Timeout as timeout_err:
307         print(f"Timeout error occurred: {timeout_err}")
308     except Exception as err:
309         print(f"An error occurred: {err}")
310 def main():
311     '''main function'''
312     owner = input("Enter the GitHub repository owner: ")
313     repo = input("Enter the GitHub repository name: ")
314     repo_info = get_repo_info(owner, repo)
315     if repo_info:
316         print(f"\nRepository: {repo_info['name']}")
317         print(f"Description: {repo_info['description']}")
318         print(f"Stars: {repo_info['stars']}")
319         print(f"Forks: {repo_info['forks']}")
320         print(f"Open Issues: {repo_info['open_issues']}")
321 if __name__ == "__main__": main()
322

```

```

Enter the GitHub repository owner: 2303A51090
Enter the GitHub repository name: AIAC_LAB

Repository: AIAC_LAB
Description: None
Enter the GitHub repository name: AIAC_LAB

Repository: AIAC_LAB
Description: None

Repository: AIAC_LAB
Description: None
Description: None
Stars: 0
Forks: 0
Open Issues: 0
PS C:\Users\arell\Music\aiac>

```

Task 4 – Real-Time Application: News Headlines Aggregator

Scenario: Build a News Aggregator using a free News API.

Requirements:

- Fetch top 5 headlines for a given category (sports, technology, health).

- Handle errors like invalid category, missing API key, and request failures.
- Display headlines in a user-friendly numbered list format.
- Include a retry mechanism if the first request fails.

Expected Output:

- A working Python script that retrieves and displays news headlines with proper error handling

SCREENSHOTS:

```

338 import requests
339 def get_news_headlines(category):
340     '''get news headlines'''
341     api_key = 'demo'
342     url = f'https://newsapi.org/v2/top-headlines?category={category}&apiKey={api_key}'
343     try:
344         response = requests.get(url)
345         response.raise_for_status() # Check for HTTP errors
346         data = response.json()
347         if data.get('status') != 'ok':
348             raise ValueError("Failed to fetch news headlines.")
349         return data.get('articles', [])
350     except requests.exceptions.HTTPError as http_err:
351         print(f"HTTP error occurred: {http_err}")
352     except requests.exceptions.ConnectionError as conn_err:
353         print(f"Connection error occurred: {conn_err}")
354     except requests.exceptions.Timeout as timeout_err:
355         print(f"Timeout error occurred: {timeout_err}")
356     except ValueError as val_err:
357         print(val_err)
358     except Exception as err:
359         print(f"An error occurred: {err}")
360 def main():
361     '''main function'''
362     category = input("Enter a news category (sports, technology, health): ")
363     headlines = get_news_headlines(category)
364     if headlines:
365         print(f"\nTop 5 headlines in '{category}':")
366         for idx, article in enumerate(headlines[:5], start=1):
367             print(f"{idx}. {article.get('title', 'N/A')}")
368             print(f"    Source: {article.get('source', {}).get('name', 'N/A')}")
369             print(f"    Published At: {article.get('publishedAt', 'N/A')}\n")
370 if __name__ == "__main__":    main()

```

Enter a news category (sports, technology, health): sports

Top 5 headlines in 'sports':

1. 2026 MLB Opening Day schedule, scores: Live updates as Mets chase Paul Skenes early in day's first game - CBS Sports
Source: CBS Sports
Published At: 2026-03-26T18:19:16Z
2. Three Keys & a Pick: No. 3 seed Illinois vs. No. 2 seed Houston - 247Sports
Source: 247Sports
Published At: 2026-03-26T17:57:29Z
3. The Final Four is in sight as the men's NCAA tournament reaches the Sweet 16. Here's what to watch on Thursday - CNN
Source: CNN
Published At: 2026-03-26T17:25:38Z
4. Storylines to watch as the Nationals' rebuild begins again - The Washington Post
Source: The Washington Post
Published At: 2026-03-26T17:00:00Z
5. Ilia Malinin leads World Championships with personal best short program - NBC Sports
Source: NBCSports.com
Published At: 2026-03-26T16:48:56Z

PS C:\Users\arell\Music\aiac> █